J. Geomag. Geoelectr., **36**, 529–563, 1984

Investigation of the Paleomagnetism of the Basement Complex of Wright Valley, Southern Victoria Land, Antarctica

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(Received July 21, 1983)

The basic magnetic properties, hysteresis, AF demagnetization, thermal demagnetization and thermo-magnetic properties of samples from five kinds of formations in the basement complex of Wright Valley (77.53°S latitude and 161.63°E longitude), Southern Victoria Land, Antarctica, have been measured. The results suggest that every sample has a stable component of natural remanent magnetization (NRM): distributed on a meridian gathered gradually to low in latitude by thermal demagnetization up to 500°C: NRM distributed in high in latitude dispersed by thermal demagnetization less than 500°C: These characteristics are related to the Curie points of the samples.

The overall results are summarized as follows. The rocks now forming the floor of Wright Valley were heated to 500°C in the Jurassic by a hidden Ferrar dolerite body. Consequently primary magnetization was remagnetized for the samples which have a Curie point lower than 500°C, but survived for the samples included in magnetite grains. Virtual geomagnetic pole (VGP) positions obtained for samples of Cambro-Ordovician age rocks consistent with previous data from rocks of the same age in East Antarctica, the VGP being situated near the equator of Africa on the present continental distribution. The differences in the declination of the Cambro-Ordovician from East Antarctica are consistent with an angular rifting of 20°–30° having occurred subsequently within East Antarctica, probably along the line of the Amery Ice Shelf and Lambert Glacier. The Transantarctic Mountains are a part of East Antarctica and the boundary between and East-West Antarctica may be located in Ross-Weddell Sea.

1. Introduction

A large, almost ice free area, on the west side on McMurdo Sound, southern Victoria Land, Antarctica, comprises three major valleys Victoria Valley in the north, Wright Valley and Taylor Valley in the south. In the basement complex of the area several kinds of dyke swarms are observed in these excavated valleys, and the sedimentary strata called the Beacon Supergroup (sandstone) overlies the basement. In most parts of the area a single sill 240 to 300 m in thickness, is intruded into the basement complex (MCKELVEY and WEBB, 1959, 1961; ALLAN and GIBSON, 1961); consequently every rock unit has the possibility of having

been reheated by dolerite sill's intrusion.

Paleomagnetic investigations have been made of samples from Wright Valley and Victoria Valley (BULL and IRVING, 1960; BULL *et al.*, 1962) and from Taylor Valley (MANZONI and NANNI, 1977). The main results of those investigations are as follows. The directions of stable components of natural remanent magnetization (NRM) in all the units of the basement complex in the Wright and Victoria Valley area (Cambro-Ordovician Age) are parallel with those in the dolerite sills (Jurassic Age). This uniformity in direction could have resulted from the geomagnetic field in this region being constant in direction for a long period in the Palaeozoic and Mesozoic, or, more likely from the reheating of the whole area during the last phase of intrusion (that of the Ferrar dolerite) in Mesozoic time. The paleomagnetic pole positions calculated from these data are in the present south Pacific Ocean. In Taylor Valley, the mean direction of NRM from four lamprophyric dykes (Cambro-Ordovician Age) was 222.6° declination and 0.6° in inclination, but other dykes and the amphybolitic basement were not fully reliable magnetically. The corresponding paleomagnetic pole position lay at 9.3°S and 26.7°E, consistent with previous results from lower Ordovician rocks from a distant area of East Antarctica. From this evidence, it appears that the basement complex in the Wright and Victoria Valleys region was heated above the Curie point, during the intrusion of the Ferrar dolerite, in the Jurassic period.

2. Geology and Geochronology

Geological investigations within the ice-free area region have been carried out in Victoria Valley (WEBB and MCKELVEY, 1959; ALLAN and GIBSON, 1961), in Wright Valley (MCKELVEY and WEBB, 1961) and in Taylor Valley (MCKELVEY and WEBB, 1959). The whole region shows essentially the same geological structure. The summarized geological sequence exposed in Wright Valley, by MCKELVEY and WEBB (1961), is shown in Table 1, and is interpreted as follows.

The basement complex consists of more than 4,500 m of folded Precambrian—Lower Cambrian marbles, hornfelses, and schists (Asgard Formation), invaded by acid plutonic rocks. The oldest intrusives are a granitic gneiss (Olympus granite) and a porphyritic granite (Dais granite). The second intrusive phase consists of microdiorite (Loke microdiorite) and granodiorite (Theseus granodiorite) dykes intruding into the Asgard Formation and Olympus and Dais granite. The third intrusive phase includes a granite (Vida granite) containing dense swarms of younger Vanda lamprophyre and porphyry dykes invading all earlier rocks. Besides these formations, dykes of a peculiar red or pink color, and 1 to 2 m in thickness, intrude into the basement complex at Bull Pass. We refer to this dyke as "red dyke" in this paper. Since the Vanda lampropyre and porphyry dykes are cut by this dyke, according to field evidence by K. Yanai, the "red dyke" represents the latest stage of intrusion.

The surface of the basement complex is overlain unconformably by more than 1,200 m of mid-Paleozoic to mid-Mesozoic sediment called Beacon

Table 1. Geochronological data of basement complex and Ferrar dolerite for McMurdo Sound, Antarctica.

Age	Group	Formation	K/Ar m.y.	Rb/Sr m.y.	U/Pb m.y.
Mesozic~ Cenozoic		McMurdo Volcanics			
		Ferrar Dolerite	147–163 (9)		
		Beacon sst			
		Kukri peneplain			
Lower Paleozoic	Victoria Intrusives	Vanda Lamprophyre			
		Vanda porphyry		470±7 (5), 480±44 (6)	
		Vida Granite	210 (2), 185 (2)	480±44 (6)	
			220 (2), 211 (2)	486±14 (7)	
			451.4±6.2 (8), 461.2±8.8 (8)		
			459.9±8.0 (8),		
Upper Cambrian~ Lower Ordvician	Wright Intrusives	Theseus Granodiorite			
		Loke Microdiorite	463.2±8.2 (8), 469.0±8.2 (8)		
			458.9±8.1 (8), 464.9±8.2 (8)		
			482.1±8.52 (8)		
		Dais Grante		500±43 (6)	
		Olympus Granite-Gneiss	520 (1)	480–495 (3)	610 (4)
				500±43 (6)	
Cambrian~ Precambrian	Skelton Group	Asgard Formation	469.1±8.1 (8)		

(1) GOLDICH *et al.* (1958), (2) MCDOUGALL (1963), (3) DEUTSCH and WEBB (1964), (4) DEUTSCH and GRÖGLER (1966), (5) JONES and FAURE (1967), (6) FAURE and JONES (1973), (7) STUCKLESS (1975), (8) MCDOUGALL and GHENT (1970), and (9) MCDOUGALL (1963).

Supergroup (sandstone). In the Olympus range, between Wright and Victoria Valleys, this basement complex and Beacon Supergroup are intruded by three sills of Jurassic Ferrar dolerite. MCKELVEY and WEBB (1961) called these sills "a", "b" and "c", from lower to higher altitude. Sill "a", 240 m thick, intrudes the basement complex: Sill "b", 180 m thick, intrudes along the boundary (peneplain) of the basement complex and Beacon Supergroup: Sill "c", 120 m thick, intrudes the Beacon Supergroup. The respective vertical distances between sill "A" and bottom of the Valley, and among the sills "a", "b" and "c" are about 600, 490 and 650 m. At the eastern end Wright Valley, relatively small volcanic cones of the Cenozoic age, called McMurdo Volcanics intruded the basement complex.

Geochronological data of the basement complex from McMurdo Sound were obtained by GOLDICH *et al.* (1958), MCDOUGALL (1963), DEUTSCH and WEBB (1964), DEUTSCH and GRÖGLER (1966), JONES and FAURE (1967), MCDOUGALL and GHENT (1970), FAURE and JONES (1973) and STUCKLESS (1975), and is summarized in Table 1. K/Ar ages of Vida granite show two distinct ranges, 185–220 m.y. and 451–461 m.y. The samples which show younger ages were collected 15 to 75 m below the contact of the lower dolerite sheet in Victoria Valley (MCDOUGALL, 1963). In general the ages given by the K/Ar method are younger than those by the Rb/Sr method. This discrepancy is explained by the basement complex being heated by the later intrusion of Ferrar dolerites, with consequent loss of Argon by diffusion. The more reliable age for the basement complex are those obtained by means of the rubidium-strontium technique: the inferred ages of the Wright and Victoria Valley intrusives are 480–500 m.y. and 470–486 m.y., respectively.

3. Sampling Sites and Samples

A total of 110 oriented blocks of rocks were collected from the Olympus granite, Theseus granodiorite, Vanda lamprophyre, Vanda porphyry and "red dyke" at an altitude of about 200 m above the bottom of Wright Valley, for paleomagnetic investigations. Most of these samples were collected along the north side of Onyx River between Vanda Station and the junction of the river and the valley of Bull Pass. Several samples were also obtained from Vida granite, 20 m below from lower boundary of sill "a" in Bull Pass. Some samples for "red dyke" were collected, 300 m below the lower boundary of sill "a", almost equidistant from the bottom of the valley and the sill boundary. Those rocks were collected, taking into consideration the distance from dykes and the sequence of intrusions. These sampling sites are shown in Fig. 1. As the inclination of geomagnetic field in this area is $-83.5°$, the direction of samples were determined by means of a sun compass, but a magnetic compass was used when it was cloudy. However, the directional error in using the magnetic compass is less than 5°. Cylindrical specimens for measurement, 1 inch both in diameter and length, were cut out from these rocks in the laboratory.

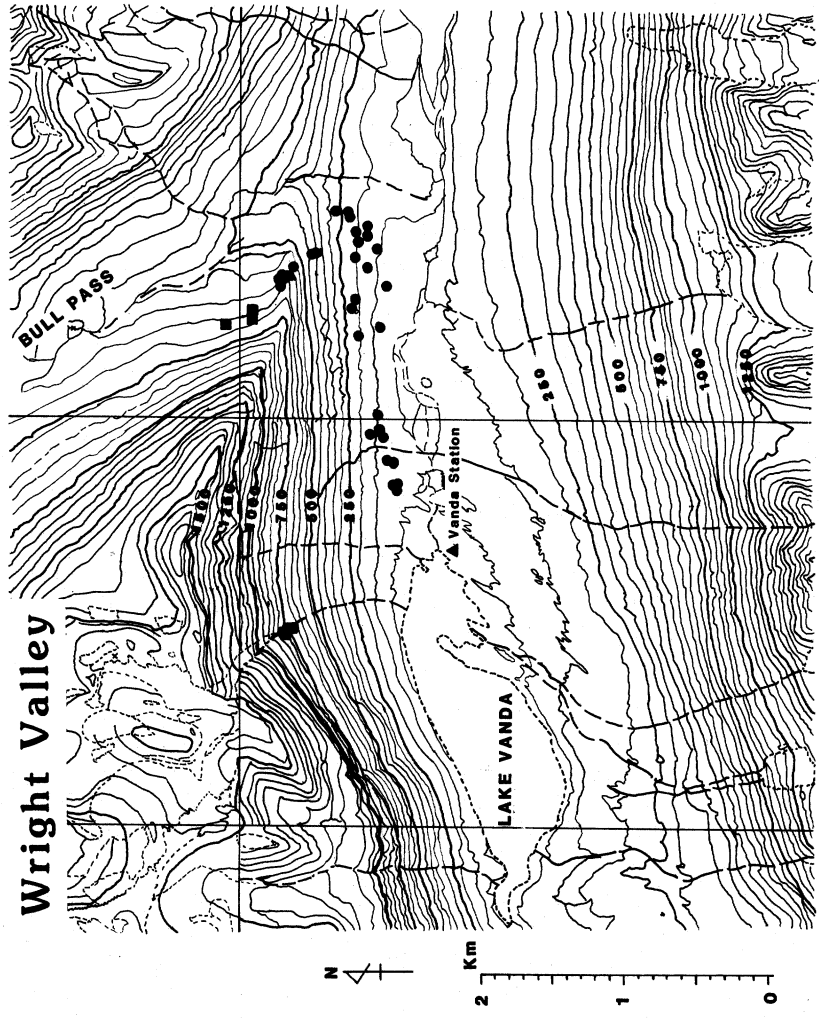


Fig. 1. Topographic map of Wright Valley, showing sampling sites for rocks of the basement complex (circle) and Ferrar dolerite (square).

4. Basic Magnetic Properties

4.1 Magnetic hysteresis properties

Magnetic hysteresis properties were determined using a vibrating sample magnetometer at room temperature, with applied magnetic fields between ±15.5 kOe. Values of the saturation magnetization (I_S), the saturation remanence magnetization (I_R), the coercive force (H_C) and the remanence coercive force (H_{RC}) are summarized in Table 2, together with the intensity of NRM (I_n) and the ratio I_n/I_S and I_R/I_S. For the granitic rocks, the Theseus granodiorite and some of the Vanda porphyry samples the value of H_{RC} could not be obtained due to the noise of the instrument.

The I_S values of the granitic rock and Theseus granodorite are weak, generally in the range $8–0.2\times10^{-3}$ emu/g, compared with those for the Vanda porphyry and "red dyke" samples which have a range of 105 to 3.5×10^{-3} emu/g. In the case of Vanda lamprophyre, samples C2 to C8 have small I_S values, $10–1\times10^{-3}$ emu/g, but the C1 sample has an extremely large value, 790×10^{-3} emu/g. The ratios I_n/I_S of all samples ranged between 2.604 and 0.192×10^{-3}, except for "red dyke" which has smaller values, 0.122 to 0.016×10^{-3}, although their ratios I_R/I_S are not very different from the others. The H_C value of granitic rocks range from 11.5 to 56 Oe. The values of Vanda lamprophyre samples have various values from 472 to 9 Oe, and those of theseus granodiorite, Vanda porphyry and "red dyke" have relatively large values, 35 to 99 Oe. These H_C values suggest that the magnetic grains in many samples are single or pseudosingledomains, but those in granitic rock, collected from the bottom of the valley, and some of the Vanda lamprophyre samples are pseudosingle or multi-domain structures. The values of H_{RC} for the Vanda lamprophyre and "red dyke" also suggest the same conclusions.

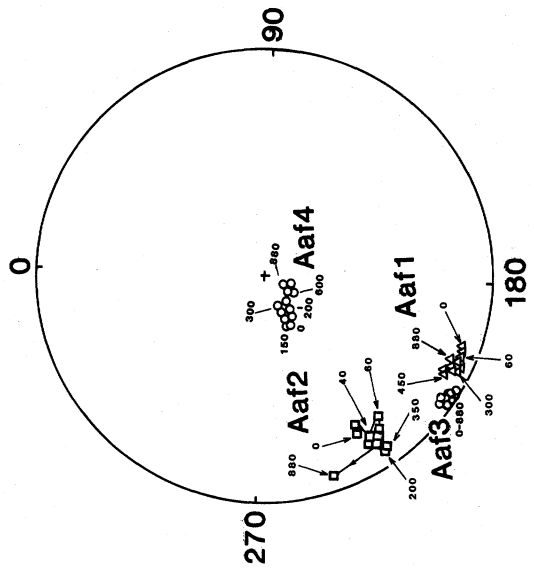
4.2 AF demagnetization

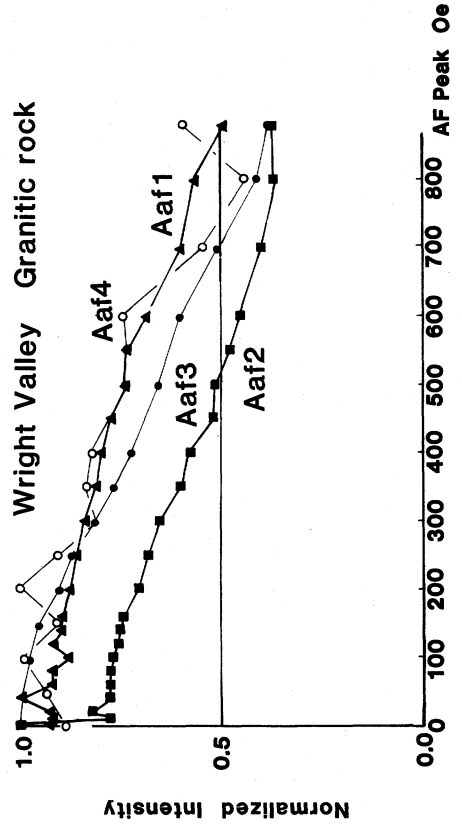
Several typical samples from each formation were demagnetized in alternating fields increased in steps up to 880 Oe peak. The AF demagnetizer consists of a two axis tumbler and a three-layer Mumetal shield case, to cancel the effect of earth's magnetic field.

The results of AF demagnetization of NRM suggest that all the samples except those from Theseus granodiorite and Vanda "red dyke" have stable NRM; they do not show a large soft magnetic component such as isothermal remanent magnetization (IRM) and viscous remanent magnetization (VRM). The median demagnetization field (MDF) of these samples exceeds 500 Oe and their directions of NRM change less than 12°, during AF demagnetization up to 600 Oe peak. Figure 2(a) shows the AF demagnetization curves of 4 samples of granitic rock, as examples of stable NRM. These representative samples were selected from the bottom of Wright Valley (Aaf 1, Aaf 2 and Aaf 3), and from 300 m above the bottom of the valley (Aaf 4). Their intensities of NRM range from 2.275×10^{-7} to 2.433×10^{-6} emu/g. The AF demagnetization curves of intensi-

Table 2. Basic magnetic properties of basement complex in Wright Valley.

Sample name		Inc	I_n	I_S	I_R	H_c	H_{RC}	I_n/I_S	I_R/I_S	I_S-T curve	
			emu/g $\times 10^{-6}$	emu/g $\times 10^{-3}$	emu/g $\times 10^{-4}$	Oe	Oe	$\times 10^{-7}$	$\times 10^{-1}$	Type	Curie point
Granitic rock	A1	−21.8	1.687	4.0	1.5	14	—	0.422	0.375		
	A2	−12.2	2.604	1.0	1.0	11.5	—	2.604	1.0		
	A3	−17.2	3.636	8.0	8.0	15.8	—	0.455	1.0		
	A4	−15.5	1.032	0.4	0.4	11.5	—	2.580	1.0		
	A5	−42.0	0.894	1.5	1.5	25	—	0.596	1.0		
	A6	−70.6	0.277	0.2	1.3	56	—	1.385	6.5		
	A7	−75.4	0.144	0.75	1.7	54	—	0.192	2.267		
Theseus grano-diorite	B1	−67	1.649	1.0	0.3	61	—	1.649	0.3		
	B2	−76	0.365	0.8	0.18	75	—	0.456	0.225		
Vanda lamprophyre	C1	−17.6	336.7	790	3630	472	688	0.426	4.595	1	570
	C2	−19.2	3.212	2	3.5	23	306	1.606	1.750	3	30–400, 570
	C3	−50.7	6.907	3	7.0	53	711	2.303	2.333	4	30–570
	C4	−41.8	2.793	2	1.3	9	146	1.397	0.650	4	30–570
	C5	−62.1	1.720	2	1.4	11	216	0.860	0.7	4	30–570
	C6	−58.5	8.393	10	40	172	1384	0.839	4.0	4	30–570
	C7	−79.1	2.123	3	4.3	28	308	0.708	1.430	5	30–400
	C8	−71.9	1.533	1	1.0	13	111	1.533	1.0	6	220, 330
Vanda porphyry	D1	−38.1	3.800	11	9.0	50	497	0.346	0.818		
	D2	−24.8	3.473	7.5	8.0	84	—	0.463	1.067		
	D3	−22.5	2.500	3.5	2.5	35	—	0.833	0.833		
Red dyke	E1	−31.8	1.663	54	34	79	471	0.0301	0.630	2	350, 560
	E2	−67.6	2.677	22	20	99	514	0.122	0.909	1	570
	E3	−60.3	1.647	105	43	51	352	0.016	0.410	1	570


90
0
180
270
Aaf1
Aaf2
Aaf3
Aaf4


Wright Valley Granitic rock
Aaf1
Aaf2
Aaf3
Aaf4
Normalized Intensity
1.0
0.5
0.0
0
100
200
300
400
500
600
700
800
AF Peak Oe

(a)

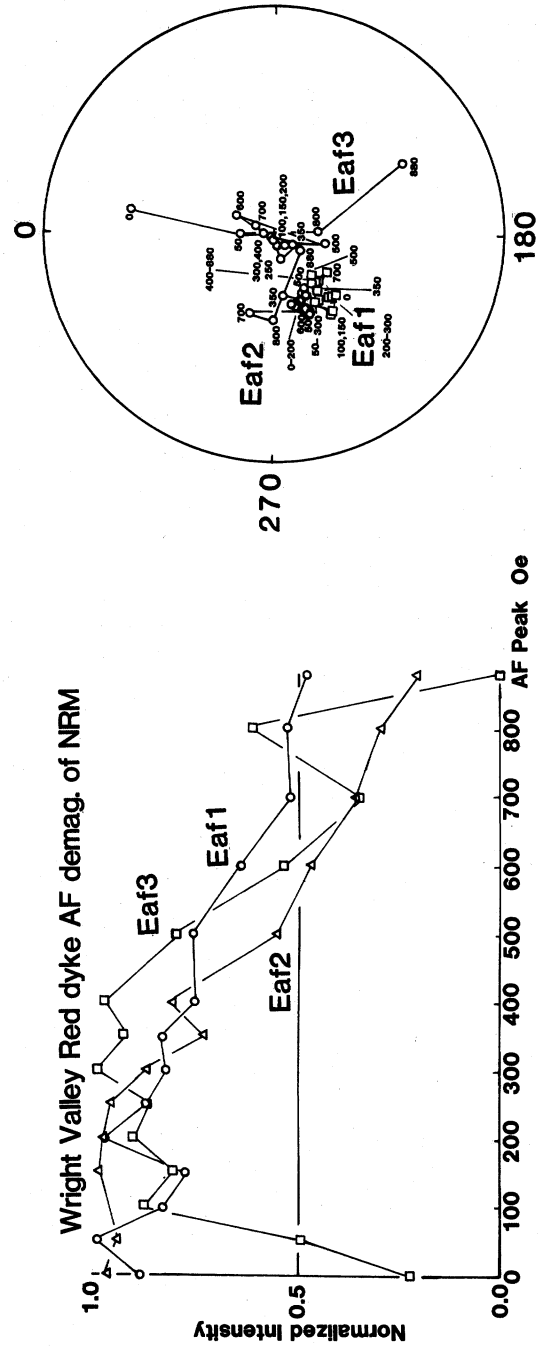


Fig. 2. AF demagnetization curves of NRM for 6 representative samples. (a): stable NRM, (b) unstable NRM.

ty, show MDF values exceeding 500 Oe peak, and are fairly stable. The directions of NRM of these samples are also stable, at least up to 800 Oe peak, as shown in Fig. 2(a). However, samples of Theseus granodiorite and "red dyke" do not show so stable a NRM against AF demagnetization. Figure 2(b) shows the AF demagnetization curves of Vanda "red dyke" samples as an example of unstable NRM. The intensities of samples Eaf 1, Eaf 2 and Eaf 3, range from 4.833×10^{-7} to 2.677×10^{-6} emu/g. These AF demagnetization curves suggest that an alternating field of about 150 Oe peak removes the unstable component from NRM, and then every samples has stable NRM against AF demagnetization, at least to 400 Oe peak.

4.3 Thermal demagnetization

Typical samples, selected according to the inclination of NRM from each formation, were thermal demagnetized in air in steps of 50 or 100°C, up to 600°C. The field intensity in the thermal demagnetizer is less than 20γ.

Because the NRM of all samples of granitic rock from the bottom of Wright Valley are distributed along a meridian from middle to low latitudes of the south hemisphere, as described later, three samples were selected from the middle and low latitude groups. The inclination of these samples ranges from $-8.5°$ to $-21.9°$ for Ath 1, Ath 2 and Ath 3 of the low latitude group, and from $-46.7°$ to $-69.0°$ for Ath 4, Ath 5 and Ath 6 from the middle latitude group. The intensities of the low latitude group (1.566–4.140×10^{-6} emu/g) decay steeply between 500° and 600°C; only one clear blocking temperature is observed between 500° and 600°C, as shown in Fig. 3(a). The directions of this group show almost no change up to 500°C and then scatter from the cluster at 600°C. For the middle latitude group, the NRMs change gradually, not only in intensity (1.034–5.797×10^{-6} emu/g) but also in direction, with thermal demagnetization up to 500°C; the directions of NRM that were in the middle latitude systematically shift to low latitudes, synchronizing with a decay of intensity up to 500°C, but they scatter widely at 600°C. Residual remanent magnetizations of random direction that are observed after heating 600°C, are due to noise in the thermal demagnetizer.

Thermal demagnetization curves of granitic rocks collected from 20 m below the lower boundary of sill "a" show that both the intensities and directions of the NRM are stable up to 400°C, but intensities decay and directions scatter above that temperature.

The results of thermal demagnetization of samples Bth 1 and Bth 2 from Theseus granodiorite are shown in Fig. 3(b). The initial intensities of Bth 1 and Bth 2 are 2.796×10^{-7} and 3.467×10^{-7} emu/g, respectively; both samples are demagnetized steeply between 300°C and 400°C. The corresponding directions, originally with high inclinations of $-79.5°$ and $-77.0°$, respectively, show almost no change up to 300° or 400°C, they then shift systematically to low latitutdes at 500°C. Again the residual remanent magnetization after heating to 600°C may be due to noise in the thermal demagnetizer.

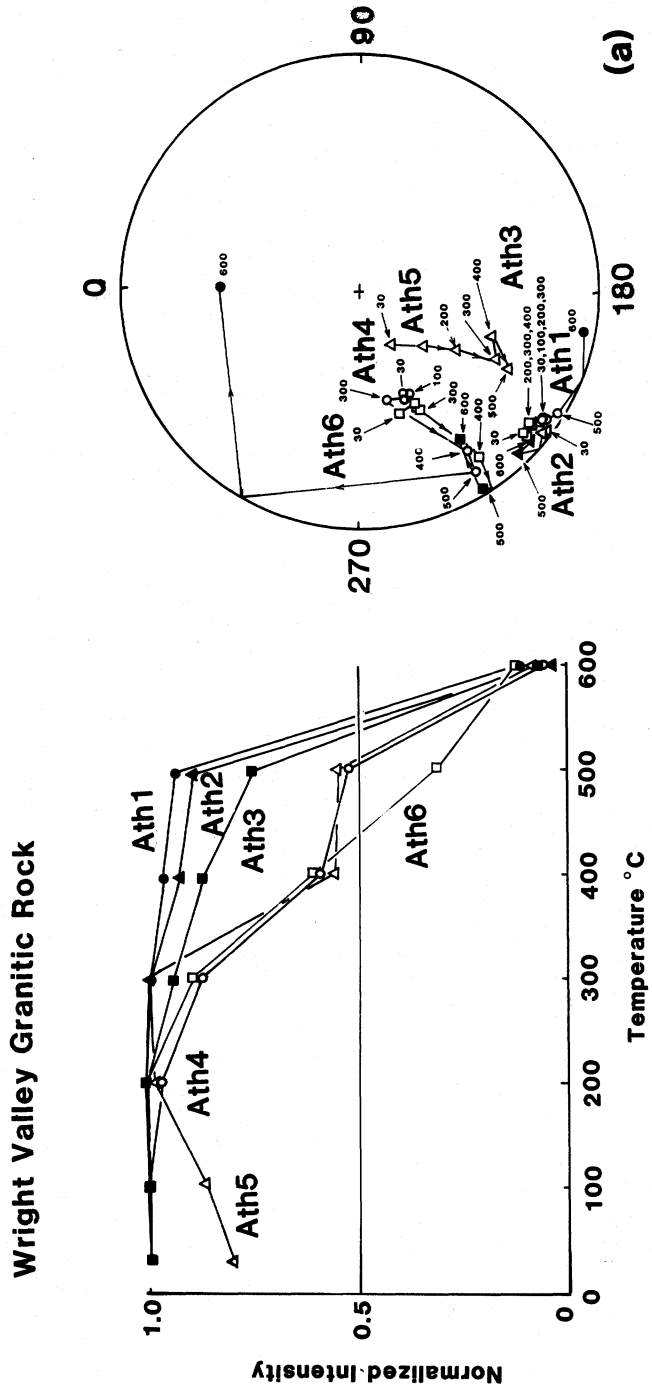
Since the NRM directions of all samples from the Vanda lamprophyre are distributed along a meridian from 0° to 90° in latitude in the southern hemisphere, two or three samples were selected from each of the high, middle and low latitude groups for thermal demagnetization. Their curves are illustrated in Figs. 3(c) and (d). Before demagnetization intensities of Cth 1 and Cth 2, of the low latitude group, were 3.431×10^{-4} and 3.933×10^{-7} emu/g, respectively. Their intensities are almost unchanged up to 300°C, but decay steeply between 500°C and 600°C as shown in Fig. 3(c). Their directions of NRM show no large changes up to 600°C, but do shift a little towards lower latitudes with increasing demagnetizing temperature. Thermal demagnetization curves of the high latitude group, Cth 6, Cth 7 and Cth 8, which are also shown in Fig. 3(c), are very similar to those of the Theseus granodiorite. The intensities, ranging from 6.201×10^{-7}, are then demagnetized steeply from 300°C to 400°C. Their inclinations of NRM, distributing from $-68.0°$ to $-84.6°$, show little change up to 300°C, but shift widely from 400°C to 600°C. Consequently it seems that the observed residual remanence at more than 400°C may be noise. In the case of the I_S-T illustrated in Fig. 3(d), the intensities, ranging among 1.1×10^{-6} to 7.7×10^{-6} emu/g, show almost no change up to 300°C, but decay gradually between 300° and 600°C. The directions of three samples have a tendency to shift to lower latitudes, simultaneously with the decay of intensities by thermal demagnetization up to 500°C. Since the directions of three samples are random at 600°C, the residual remanences are probably noise. These I_S-T curves are, therefore, fairly similar to those of the "middle latitude" group of granitic rocks collected from the bottom of Wright Valley.

The thermomagnetic curves of Dth 1, Dth 2 and Dth 3, samples of Vanda porphyry, which are shown in Fig. 3(e), are fairly similar to those of the "low latitude" group of granitic rocks from the bottom of Wright Valley, and the Vanda lamprophyre. Their original intensities of NRM, which ranged among 3.463 to 3.723×10^{-6} emu/g, do not decay very much up to 500°C and then decay steeply from 500° to 600°C. Their directions of NRM, remain at low inclinations, with almost no change, up to 400°C, move to even lower latitudes at higher temperatures and start to scatter at 600°C.

NRMs of samples Eth 1, Eth 2 and Eth 3, of the Vanda "red dyke" are unstable against thermal demagnetization, compared with those of other formations, as shown in Fig. 3(f). Eth 1 and Eth 3 were collected from the bottom of the Wright Valley while Eth 2 was collected from 300 m above the bottom of the valley. Their NRM intensities, which initially ranged among 4.833×10^{-7} to 2.677×10^{-6} emu/g, decay smoothly from 200° to 600°C. The directions of NRM are relatively stable from 300° to 450°C for Eth 1 and from 100° to 450°C for Eth 2 and Eth 3.

4.4 Thermomagnetic curves

The thermomagnetic curves (I_S-T curve) were obtained using a vibrating sample magnetometer, at 1×10^{-4} torr atmospheric pressure, from room temperature


Wright Valley Granitic Rock
Normalized Intensity
1.0
0.5
0
0
100
200
300
400
500
600
Temperature °C
Ath1
Ath2
Ath3
Ath4
Ath5
Ath6
(a)
90
180
270
0

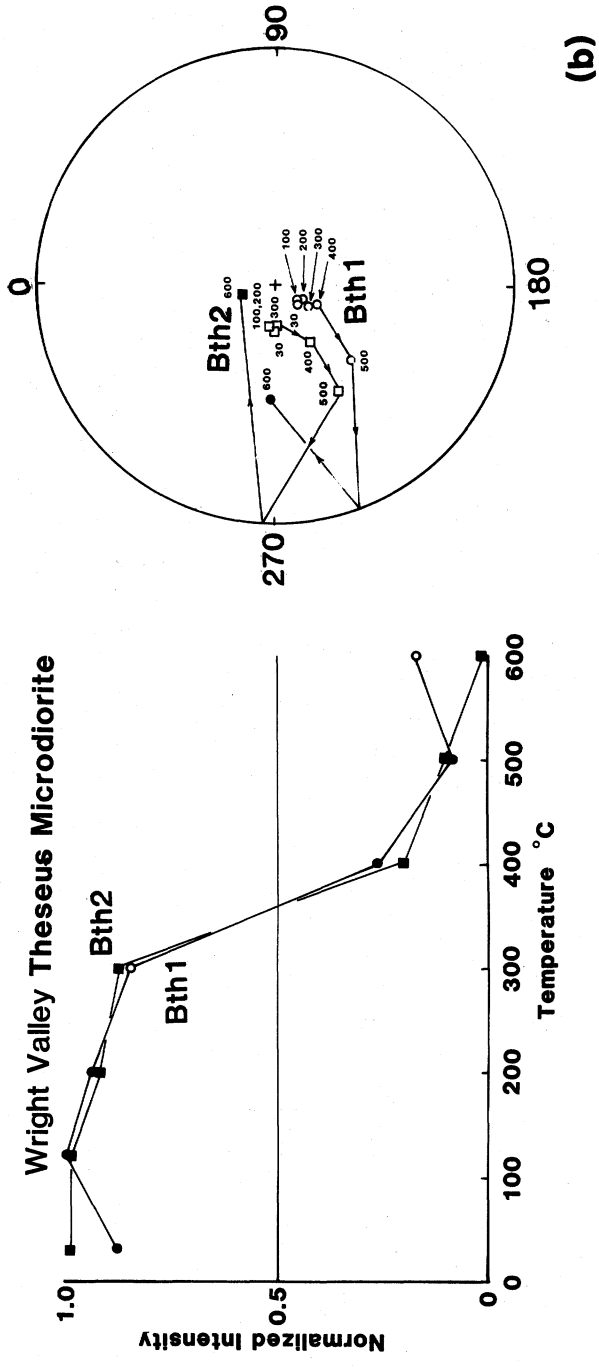
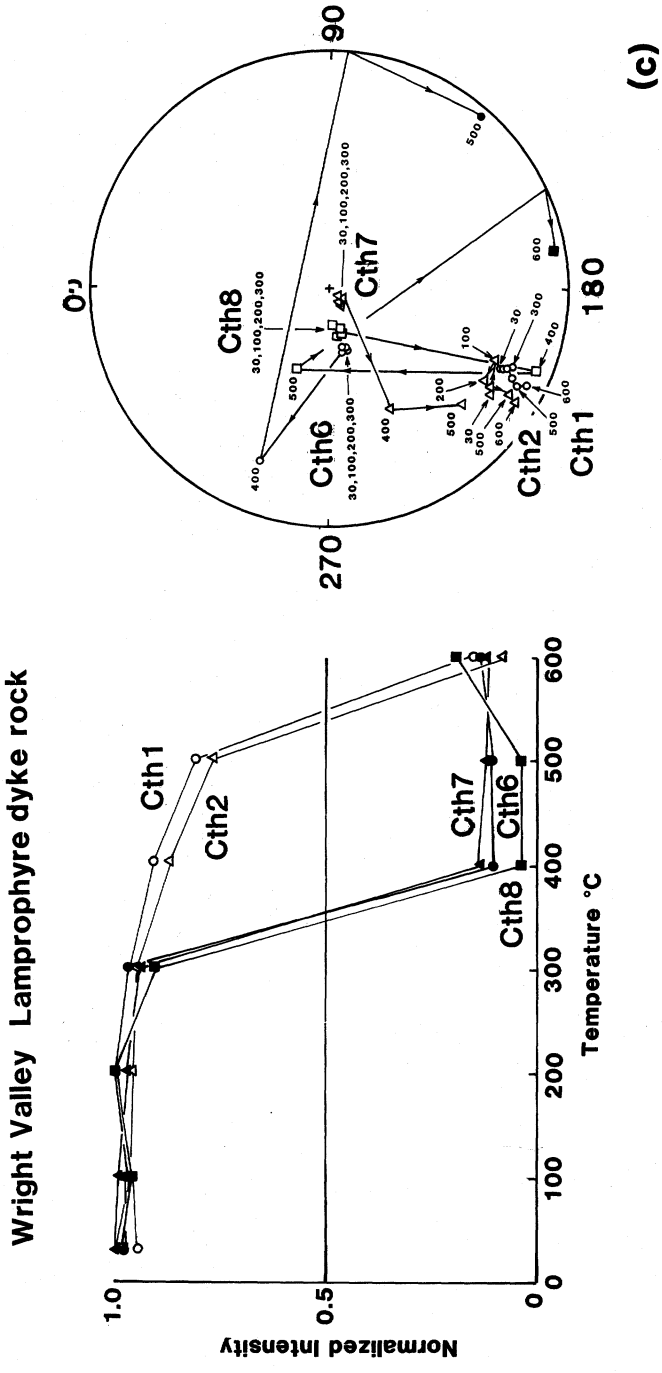

(b)
90
0
180
270
Bth2
Bth1
600
800
100,200
300
30
400
500
100
200
300
400
Wright Valley Theseus Microdiorite
Bth2
Bth1
Normalized Intensity
1.0
0.5
0
0
100
200
300
400
500
600
Temperature °C

Fig. 3.


Wright Valley Lamprophyre dyke rock
Normalized Intensity
1.0
0.5
0
0
100
200
300
400
500
600
Temperature °C
Cth1
Cth2
Cth7
Cth6
Cth8
(c)
90
180
270
0
Cth8
30,100,200,300
Cth7
Cth6
Cth2
Cth1

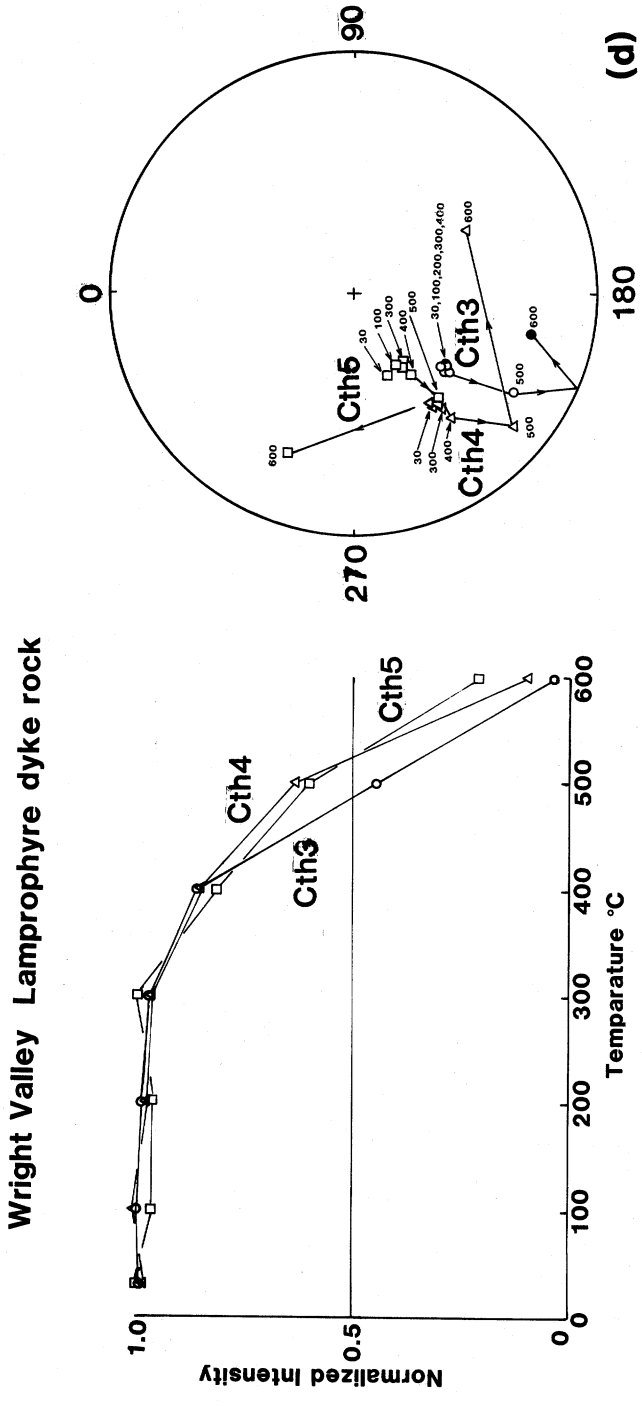
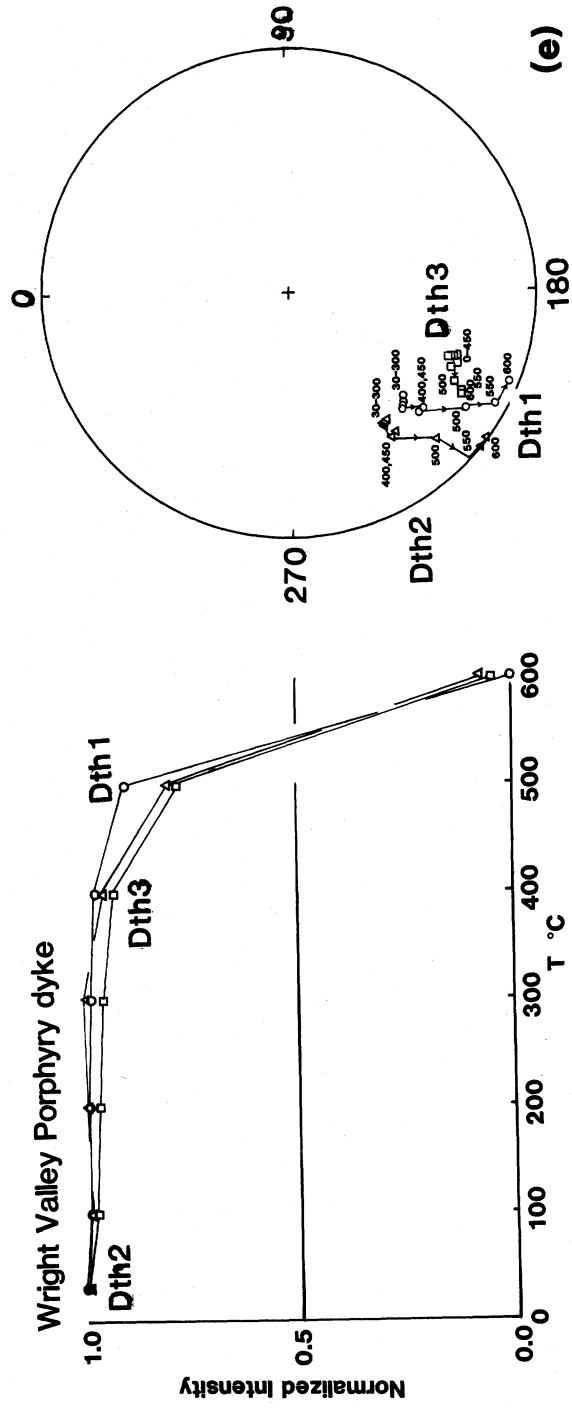

Wright Valley Lamprophyre dyke rock
Normalized Intensity
1.0
0.5
0
0
100
200
300
400
500
600
Temparature °C
Cth3
Cth4
Cth5
(d)
90
180
270
0
Cth5
Cth3
Cth4
30
100
300
400
500
30,100,200,300,400
600
600
500
500
30
300
400

Fig. 3.


(e)
90
0
180
270
Dth1
Dth2
Dth3
Wright Valley Porphyry dyke
Normalized Intensity
1.0
0.5
0.0
0
100
200
300
400
500
600
T °C

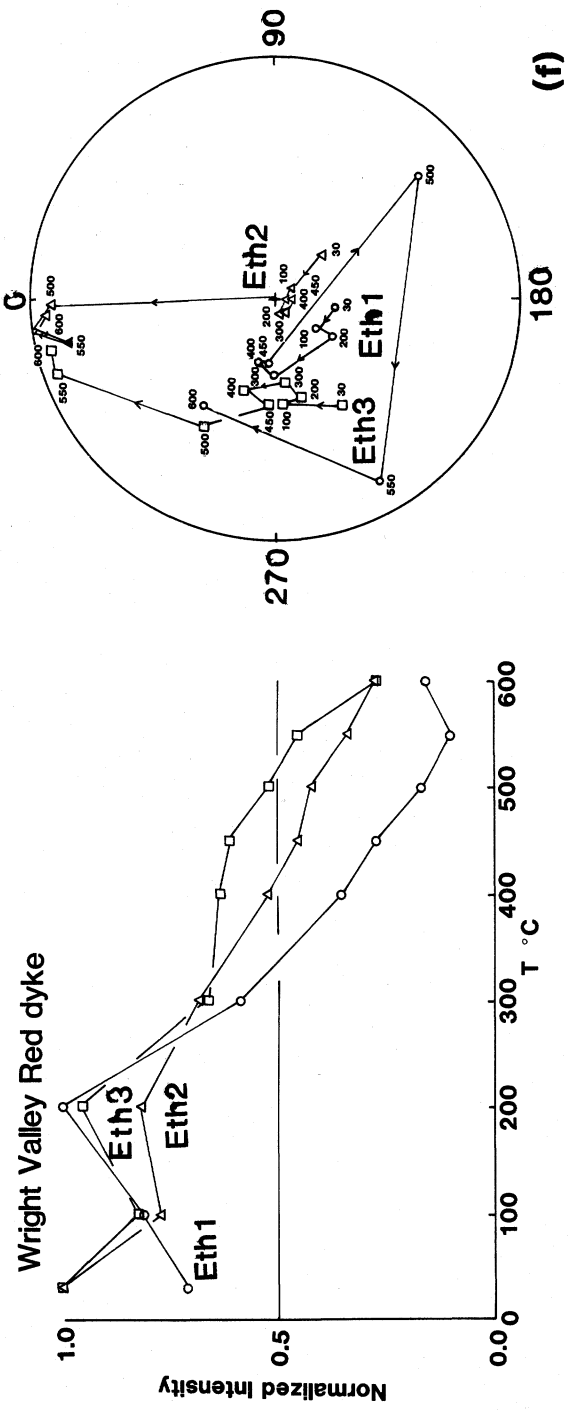


Fig. 3. Thermal demagnetization curves of representative samples of basement complex. (a): granitic rocks, (b): Theseus granodiorite, (c) and (d): Vanda lamprophyre, (e): Vanda porphyry, (f): "red dyke".

to 650°C. The applied field intensity is 5K Gauss and heating and cooling rates are 200°C/hour. I_S-T curves from granitic rock, Theseus granodiorite and Vanda porphyry could not be obtained because of instrumental noise. I_S-T curves obtained from typical samples of Vanda lamprophyre and "red dyke" are classified into six types, as shown in Fig. 4. Type 1 is irreversible, with a single Curie point at 570°C; ferromagnetic minerals are magnetite of constant composition, but they may have a low temperature oxidation (titanomagphemite), because of the increased magnetization produced during the heating. Type 2 is irreversible, with tow distinct Curie points at 350°C and 570°C in the heating curve and at 570°C in cooling curve. Since the Curie point at 350°C has disappeared in the cooling curve, it is concluded that a part of the magnetic grains change to titanomagphemite. Type 3 is reversible with no well-defined Curie points except the obvious one at 570°C. Titanomagnetite grains, with various compositions, and magnetite are inferred. Type 4 is reversible, with minor Curie points at about 300°, 380° and 370°C. Titanomagnetite grains with varying composition and a small quantity of magnetite are inferred. Type 5 is reversible, with no well-defined Curie point. Titanomagnetite with varying compositions are inferred. Type 6 is irreversible with one Curie point at 220°C, in the heating curve, and no well-defined Curie points in the cooling curve, are observed. Titanomagnetite may be chemical altered during the heating. The classification of each sample into these types is shown in Table 2. A relationship between the inclination of NRM and Curie point for these samples is recognized in the samples of Vanda lamprophyre: samples of Type 1 and 3, which show the distinct Curie point of magnetite, have low inclination NRM, at about $-20°$; those of Type 5 and 6, which have lower Curie points less than 500°C, have high inclinations of $-70°$ to $-80°$; those of Type 4 lie between the two other cases. However, there is no apparent correlation between Curie points and inclinations for Vanda lamprophyre. Pilot samples from "red dyke", which all show Type 2 I_S-T curves, have NRM inclinations of $-32°$ to $-65°$.

5. Mean Direction of NRM against Demagnetizations

The NRMs of every sample, before and after AF demagnetization in magnetic fields of up to 140 Oe (peak), were measured, and then these samples were thermal demagnetized at 300°, 400°, 500°, and 550°C in air, based on the results of thermal demagnetization of representative samples. The results obtained by means of these demagnetizations are summarized in Table 3 and are plotted in Fig. 5(a) to (d). Table 3 shows the number of examined specimens (N) collected from the same formation, the mean intensity of NRM (R), the mean inclination (Inc) and declination (Dec) of NRM, the estimate of precision (K), the semiangle of the cone of confidence of 95% probability (α_{95}), and the paleolatitude (pLat) and paleolongitude (pLon) of virtual geomanetic pole (VGP).

The individual NRM directions of granitic rock from the bottom of the Wright Valley, plotted after every demagnetization step, are shown in Fig. 5(a),

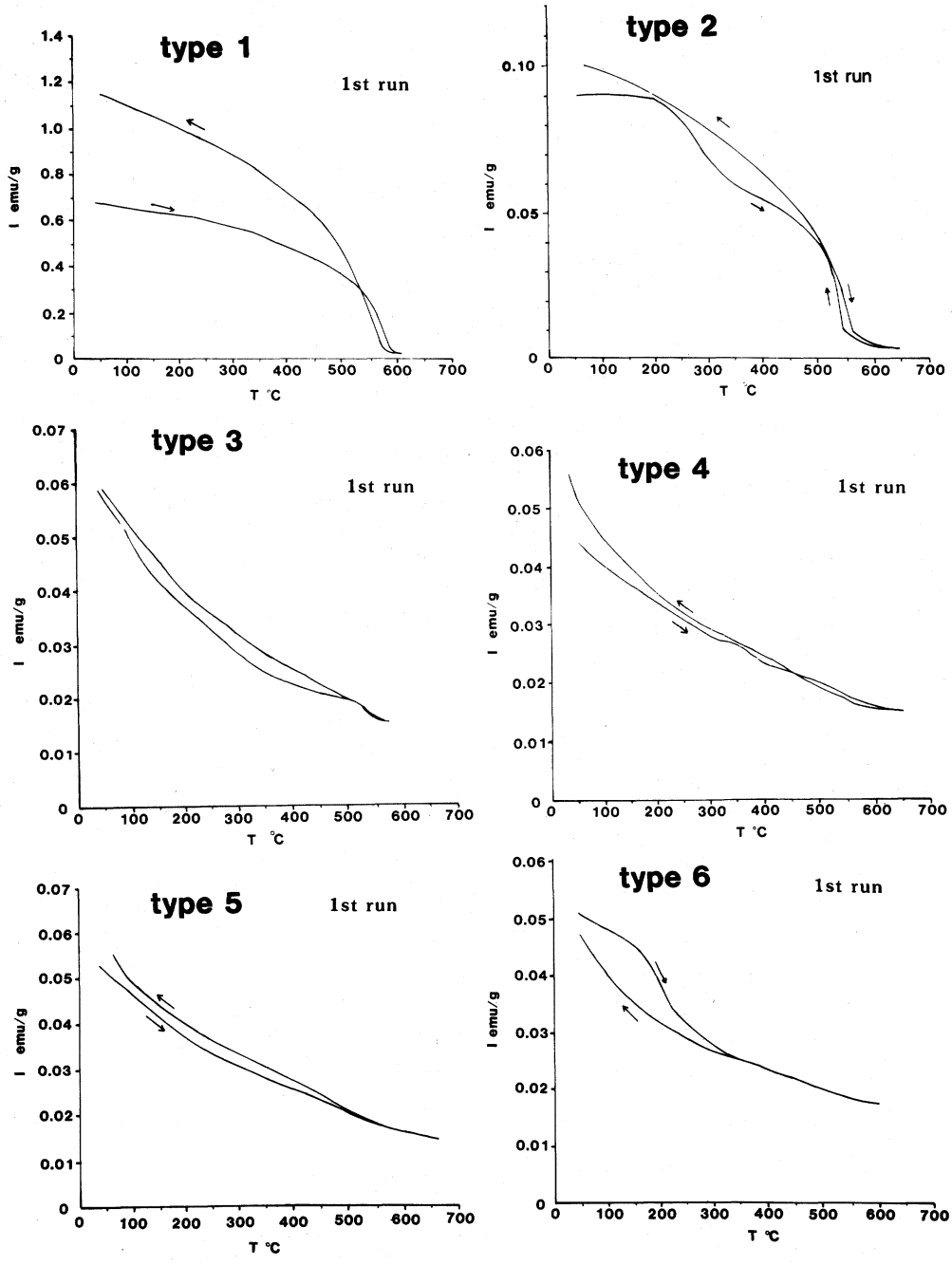


Fig. 4. Classification into 6 types of the thermomagnetic curves for Vanda lamprophyre and "red dyke" samples.

Table 3. Change of the directions of NRM by AF and thermal demagnetizations.

Sample name			NRM	140° Oe	300°C	400°C	500°C	550°C
A Wright Valley granitic rocks	Bottom of valley	N	16	16	16	16	16	9
		R emu/g	20.9×10^{-6}	2.02×10^{-6}	1.89×10^{-6}	1.77×10^{-6}	1.67×10^{-6}	1.40×10^{-6}
		Inc	−26.5	−26.9	−24.3	−13.5	−9.0	−2.7
		Dec	220.5	219.8	218.7	215.9	216.7	217.4
		K	17.8	17.3	17.7	20.6	21.5	24.9
		α_{95}	8.7	8.8	9.0	8.3	8.1	10.5
		pLat	4.4N	4.6N	3.0N	3.3N	5.4N	8.5N
		pLon	20.9N	20.2E	19.4N	17.4N	18.5N	19.6E
	20 m below from bottom of Valley	N	4	4	3	3	3	
		R emu/g	1.40×10^{-7}	1.384×10^{-7}	1.17×10^{-7}	8.87×10^{-7}	1.85×10^{-7}	
		Inc	−69.1	−73.4	−70.9	−73.2	−9.4	
		Dec	237.7	237.0	222.7	217.8	152.6	
		K	376.2	247.6	218.7	30.3	2.2	
		α_{75}	4.7	5.9	8.4	22.8	122.8	
		pLat	44.9S	51.1S	45.5S	48.5S	6.3S	
		pLon	151.8W	155.1W	164.8W	169.8W	134.0E	
B Theseus granodiorite		N	10	10	10	10	10	
		R emu/g	2.63×10^{-7}	2.43×10^{-7}	2.29×10^{-7}	9.13×10^{-7}	1.11×10^{-7}	
		Inc	−75.5	−76.8	−75.8	−63.6	−24.3	
		Dec	255.3	255.3	230.2	221.5	252.4	
		K	84.0	105.8	130.2	22.4	2.4	
		α_{95}	5.3	4.7	4.3	10.4	41.0	
		pLat	54.4S	59.5S	54.0S	35.5S	8.7S	
		pLon	142.7W	144.4W	162.0W	163.3W	128.1W	

Sample name		NRM	140° Oe	300°C	400°C	500°C	550°C
C Vanda	N	39	39	38	36	31	29
lamprophyre	R emu/g	1.06×10^{-4}	1.05×10^{-4}	9.94×10^{-5}	8.96×10^{-5}	9.29×10^{-7}	4.46×10^{-5}
	Inc	−37.5	−35.3	−36.8	−30.3	−20.3	17.4
	Dec	228.1	228.6	224.4	218.3	219.8	210.0
	K	4.3	3.9	5.2	12.3	15.0	1.9
	α_{95}	12.6	13.5	10.5	7.1	6.9	28.5
	pLat	12.5N	11.2N	11.5N	6.3N	0.9N	19.6N
	pLon	25.7E	26.5E	23.7E	18.6E	20.8E	13.4E
D Vanda	N	28	28	27	27	27	25
porphyry	R emu/g	3.59×10^{-6}	3.57×10^{-6}	2.85×10^{-6}	2.76×10^{-6}	2.39×10^{-6}	1.57×10^{-6}
	Inc	−25.1	−26.8	−24.3	−21.7	−6.6	24.1
	Dec	227.8	228.6	224.5	222.4	223.0	236.8
	K	39.1	42.0	38.2	41.9	13.0	3.0
	α_{95}	4.4	4.3	4.6	4.4	8.0	20.3
	pLat	4.7N	5.8N	3.8N	2.0N	5.8N	19.2S
	pLon	28.1E	28.8E	25.0E	23.2E	24.9E	41.5E
E Vanda	N	13	13	13	13	13	
"red dykes"	R emu/g	2.34×10^{-6}	$2.05-10^{-6}$	1.93×10^{-6}	1.45×10^{-6}	1.20×10^{-6}	
	Inc	−53.4	61.7	−59.7	−66.8	−53.0	
	Dec	238.1	238.4	260.6	252.6	269.9	
	K	6.9	22.0	26.5	22.5	4.7	
	α_{95}	17.7	9.0	8.2	8.9	21.6	
	pLat	26.9S	35.5S	37.5S	38.3S	32.7S	
	pLon	146.2W	141.8W	127.5W	135.6W	116.5W	

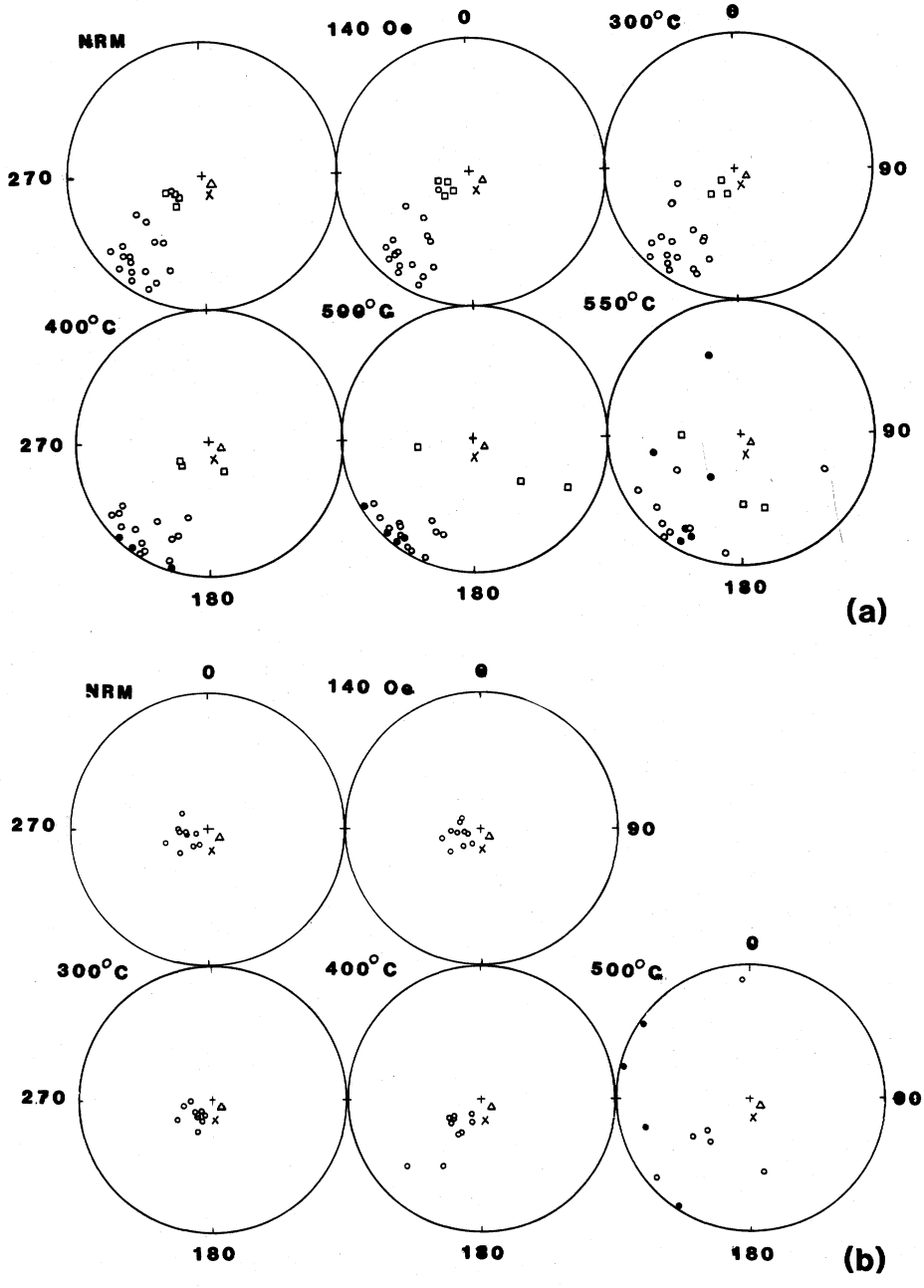
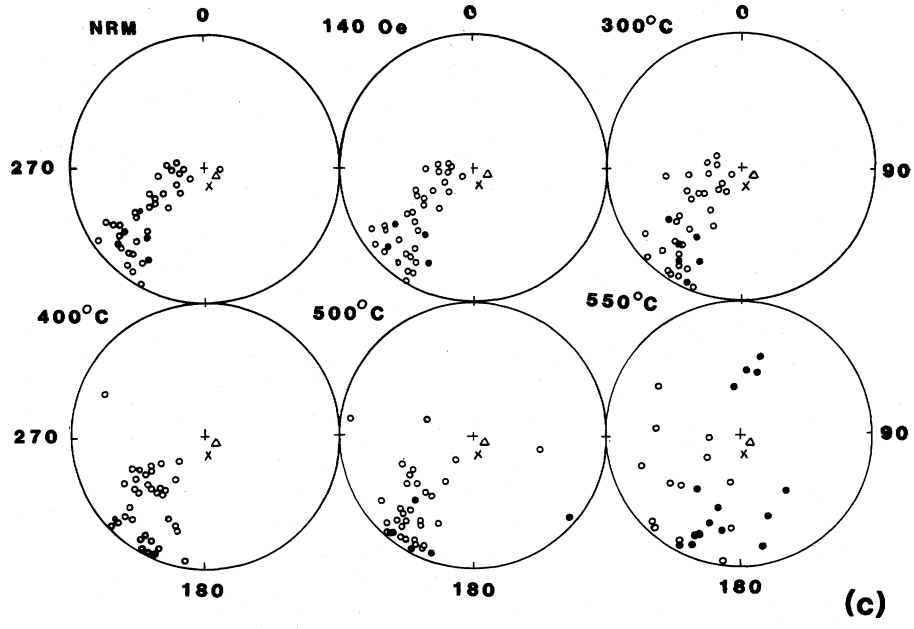
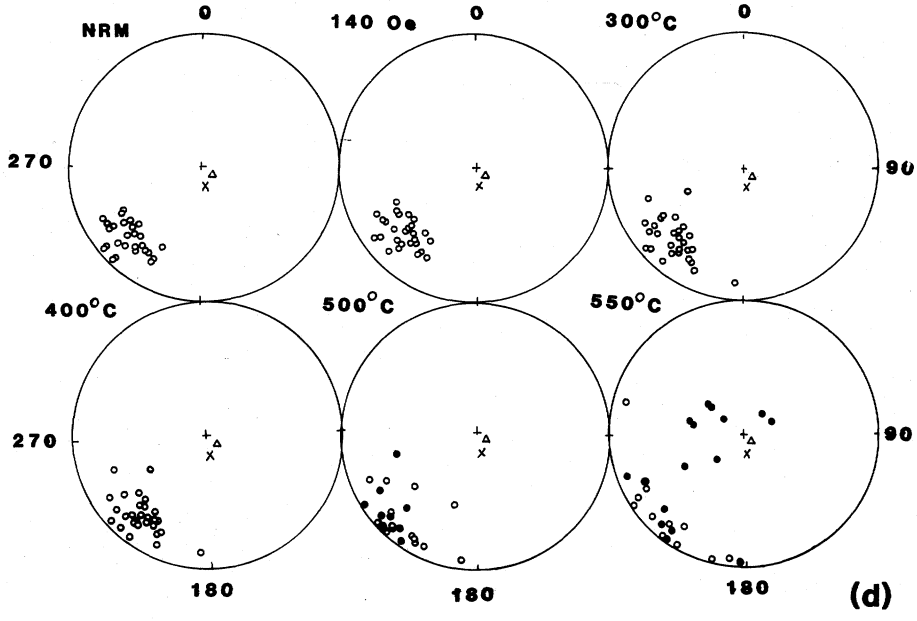


Fig. 5. Change of the directional NRMs after AF demagnetization to 140 Oe and thermal demagnetization to 300°, 400°, 500° and 550°C. Triangle: direction of geomagnetic dipole at present; cross: geomagnetic field direction for Wright Valley; open circle and square: normal polarity; solid





circle: reversed polarity. Equal area projection. (a): granitic rock, (b): Theseus granodiorite, (c): Vanda lamprophyre, (d): Vanda porphyry.

symbolized with a circle, together with direction of Earth's geomagnetic field, marked x. These specimens have directions of magnetization significantly different from the Earth's geomagnetic field in the McMurdo Sound. Directions of NRM for most specimens have low to middle latitudes in the southern hemisphere, although one is located high in latitude. This distribution pattern is not changed by AF demagnetization up to 140 Oe peak. However those directions distributed in the middle to high latitudes tend to shift towards low latitudes after thermal demagnetization, as shown in Fig. 5(a). The individual directions after thermal demagnetization at 500°C lie along a great circle of low inclinations, quite close to the horizontal plane. At 550°C the directions of 9 specimens have a cluster at the same position as those at 500°C but the others scatter from the cluster. A remarkable feature of the mean direction of NRM, during thermal demagnetization up to 550°C is the decrease of inclination for these 9 clustered specimens from $-26.5°$ to $-2.7°$, while the declination changes only from 220.5° to 216.7°, as shown in Table 3. The confidence angle α_{95}, has a minimum value at 500°C, of 8.1°. Taking into consideration the number of specimens thermal demagnetization at 500°C has the most significant clustering for NRM directions of the granitic rock from bottom of the Wright Valley, Mean difference of directional magnetization between original NRM and NRM after thermal demagnetization to 500°C is calculated as Inc $= -66.6°$ and Dec $= 274.7°$, although the value of confidence is small with $\alpha_{95} = 24.5°$. This direction is consistent to that of the Ferrar dolerites in the Wright Valley, Inc $= -69.4°$, Dec $= 237.6°$ and $\alpha_{95} = 2.4°$ (FUNAKI, 1982).

The directions of the original NRM of granitic rock, symbolized with a square, collected 20 m below the lower boundary of dolerite sill "a", has a cluster, with confidence $\alpha_{95} = 4.7°$, but they scatter by thermal demagnetization as shown in Fig. 5(a). The best precision and confidence are observed at original NRM. The inclination and declination of the original mean NRM direction are Inc $= 69.1°$ and Dec $= 237.7°$, respectively with confidence of $\alpha_{95} = 4.7°$ (see Table 3). This direction is fairly consistent with that of sill "a".

Individual NRM's directions of specimens from Theseus granodiorite cluster around an inclination of $-75.5°$ and a declination of 255.3°(Fig. 5(b)), with confidence of $\alpha_{95} = 5.3°$, as shown in Table 3. These original directions are further clustered by AF demagnetization at 140 Oe peak and thermal demagnetization at 300°C, but are scattered by those at 400°C and 500°C. The best clustering is observed after thermal demagnetization at 300°C. These inclination and declination of the mean NRM at that temperature are $-75.8°$ and 230.2° respectively, with a confidence of $\alpha_{95} = 4.3$. As the angular deviation between this direction and that of sill "a" is only 6.8°, taking into account the α_{95} values, the two directions are in reasonably good agreement with each other.

The original NRM directions of specimens from the Vanda lamprophyre are distributed from high to low latitude along a mean meridian of 228.1°, as shown in Fig. 5(c). The patterns of changes of direction and intensity are quite different from dyke to dyke. In general only small changes are produced by

AF demagnetization at 140 Oe peak and by thermal demagnetization at 300°C, and α_{95} ranges from 10.5° to 13.5°. However these directions are considerably clustered by thermal demagnetization at 400°C and 500°C, although a few of them scatter from the cluster. The high and middle inclinations are decreased notably at those temperatures. This behavior with thermal demagnetization is similar to that of the granitic rock from the bottom of Wright Valley. After thermal demagnetization at 550°C, the NRM directions become random. In Table 3 the data of Vanda lamprophyre samples that deviated from a cluster after thermal demagnetization at 400°C and 500°C were discarded, for statistical calculations. Since the best precision and confidence, showing $K=15.5$ and $\alpha_{95}=6.9$, are obtained by thermal demagnetization at 500°C, the inclination of $-20.3°$ and declination of 219.8°, for that temperature are considered the most significant paleomagnetic directions for the Vanda lamprophyre.

The distributions of NRM directions from Vanda porphyry samples are shown in Fig. 5(b). All specimens have middle to low inclination for their initial magnetization, this direction of magnetization being significantly from the Earth's geomagnetic field, and they are well clustered, with $K=39.1$ and a confidence of $\alpha_{95}=4.4°$. This distribution is not changed by AF demagnetization at 140 Oe peak. However, it is changed by thermal demagnetization at 300°C to 500°C; the mean direction shifts to lower inclinations, with most sample directions remaining in a cluster. A clear cluster still lies on the equator after demagnetization at 550°C, but the directions of many specimens are further scattered from their position at 500°C. A total of 9 specimens showing reverse magnetization have inclinations that tend to cluster at high inclinations, as shown in Fig. 5(d), but the mean direction of those points is consistent with the magnetic field direction in the thermal demagnetizer and these intensities are weak, less than 30% of their original value. These directions therefore are probably noise, and have been discarded for the statistical calculation in Table 3. The original mean NRM direction is an inclination of $-25.1°$ and a direction of 227.8°, these become $-6.6°$ and 223.0°, respectively, after thermal demagnetization at 500°C.

With Vanda "red dyke" samples, the initial directions of NRM, magnetized normal polarity, are not well clustered. However, NRM after AF demagnetization at 140 Oe, are clustered with a mean inclination of $-61.7°$ and a declination of 238.4°, with α_{95} of 8.1°, as shown in Table 3. The clustering is not changed with thermal demagnetization at 300°C and 400°C, but these directions are dispersed by thermal demagnetization at more than 500°C.

This demagnetization may be summarized as follows. The mean NRM directions of each formations, except for the "red dyke" rocks, are not changed appreciably by AF demagnetization at 140 Oe peak. With specimens from "red dyke" rocks, individual NRM directions are clustered by AF demagnetization. The mean NRM directions of granitic rocks collected from the bottom of the Wright Valley, Vanda lamprophyre and porphyre are all gradually shifted to lower inclinations by thermal demagnetization, and settle along a 210°–230° meridian, and low inclinations at 500°–550°C. The differential directional NRM

between the original mean NRM and that after thermal demagnetization at 500°–550°C is fairly similar to that of Ferrar dolerite. The specimens collected 20 m below the lower boundary of dolerite sill "a", from the Theseus granodiorite and from the "red dyke" have a stable NRM direction, similar to that of the Ferrar dolerite. Namely they have both or the other one of the NRM direction of Ferrar dolerite and horizontal direction.

From the thermal demagnetization results of all specimens and basic magnetic properties, the magnetization mechanism for rock formations of the basement complex in the Wright Valley may be explained as follows. The ambient geomagnetic field when the basement complex was formed, 480–500 m.y. ago (lower Ordovician to early Cambrian) was almost horizontal in this area. The region was heated by intrusion of dolerite sills about 160 m.y. ago (Jurassic Age), up to 500°C at the bottom of the Wright Valley. Consequently the NRM directions of those specimens magnetized in Cambro-Ordovician Age, which have blocking temperatures higher than 500°C, have survived (primary magnetization). However the primary magnetizations of the specimens, which have blocking temperatures lower than 500°C, have completely disappeared, and the specimens have been remagnetized (secondary magnetization) in the direction of the geomagnetic field when the dolerite sills intruded into of the area during the Jurassic. On the other hand, before thermal demagnetization those specimens which include both magnetite and various compositions of titanomagnetite show a superposition of both the primary and secondary magnetization directions. The results suggest that granitic rocks collected from 20 m below the lower boundary a dolerite sill and the Theseus granodiorite have a main blocking temperature below 400°C and shown mainly the secondary magnetization: granitic rocks collected from the bottom of the valley and the Vanda lamprophyre have various blocking temperatures and the initial NRM directions are distributed between the primary and secondary magnetization directions: Vanda porphyry has a main blocking temperature higher than 500°C and shows mainly the primary magnetization. With the Vanda "red dyke", the directions of the stable component of magnetization are similar to those of the secondary magnetization, although it shows the Curie point at 300°, 380° and 570°C. The thermal demagnetization results of these samples may suggest that the main blocking temperature depends on the 300°C and 380°C of Curie point rather than the 570°C one.

6. Paleomagnetic Discussions

The VGP positions obtained from mean NRM directions of each formations or sampling sites are shown in Table 4 and are illustrated in Fig. 6. Rb/Sr ages are taken as 500 ± 43 m.y. (FAURE and JONES, 1973) for granitic rock and as 470 ± 7 and 480 ± 44 (JONES and FAURE, 1967; FAURE and JONES, 1973) for the Vanda porphyry. The field evidence of the intrusive sequence shows that the Vanda lamprophyre is younger that the granitic rocks but older than the Vanda porphyry (MCKELVEY and WEBB, 1961). Therefore there is a possibility that the

Table 4. Paleomagnetic results of the Cambro-Ordovician rocks and Ferrar dolerite for East Antarctica.

	Sampling Site	N	Inc	Dec	α_{95}	pLat	pLon	Remarks
1.	Ongul Island	18	51	337	7	19S	17E	NAGATA and SHIMIZU (1959)
2.	Ongul Island	5	49	350	—	9S	32E	NAGATA and YAMA-AI (1961)
3.	Lützow-Holm bay	7	60	341	—	21S	24E	NAGATA and YAMA-AI (1961)
4.	Lützow-Holm bay	4	46.5	353	—	7S	33E	KANEOKA *et al.* (1968)
5.	Sør Rondane Mts.	16	64	341.5	4.5	28.5S	9.5E	ZIJDERVELD (1968)
6.	Mirnyy Station	37	16.9	294.2	15.6	1.5S	28.5M	MCQUEEN *et al.* (1972)
7.	Taylor Valley	51	0.6	222.6	10.9	17S	21E	MANZONI and NANNI (1977)
8.	Wright Valley	16	−9.0	216.7	8.1	5.4S	18.5E	This study. Granitic rocks from bottom of Wright Valley. Thermal demag. at 500°C.
9.	Wright Valley	58	−13.2	222.4	5.2	2.5S	23.8E	This study. Vanda lamprophyre and porphyry from Wright Valley. Thermal demag at 500°C.
10.	Wright Valley	13	−61.7	238.4	9.0	35.5S	148.1W	This study. Vanda "red dyke." AF demag. at 140 Oe.
11.	Wright Valley	4	−69.1	237.7	4.7	44.9S	151.8W	This study. Granitic rocks from 20 m below of lower boundary of Ferrar dolerite.
12.	Wright Valley	10	−75.8	230.2	4.3	54.0S	162.0W	This study. Theseus Granodiorite. Thermal demag. at 300°C.
13.	Wright & Victoria Valley	58	−64	247	4	40S	140W	BULL *et al.*
14.	Wright Valley (Ferrar dolerite)	26	−69.4	237.6	2.4	45.3S	152W	This study.
15.	Wright & Victoria Valleys (Ferrar dolerite)	83	−68	250	3	45S	140W	BULL *at al.* (1962)

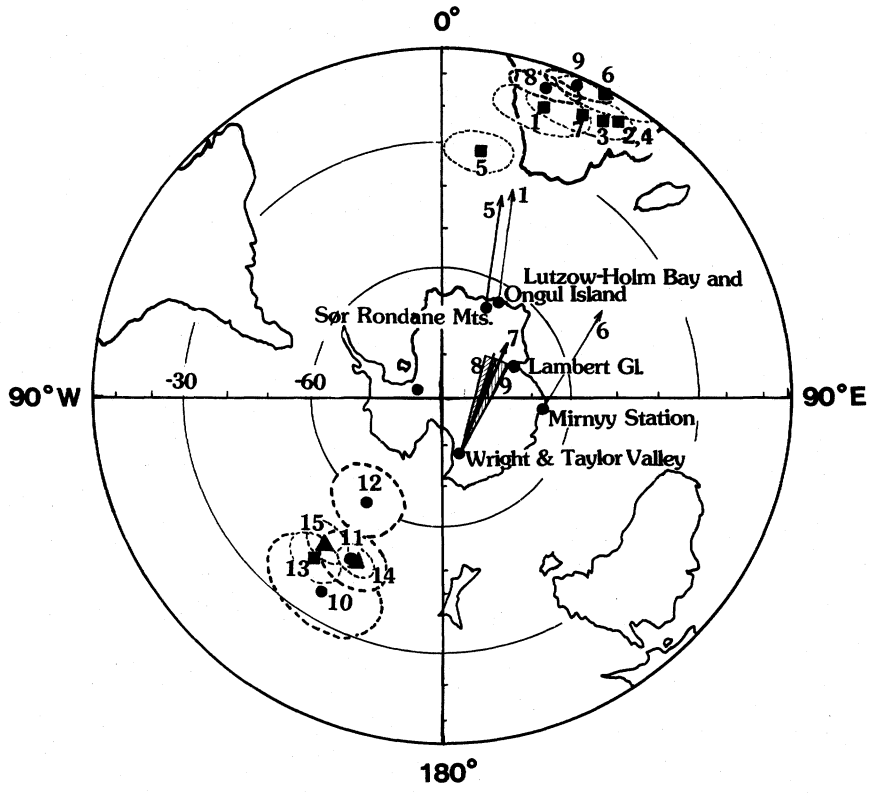


Fig. 6. Positions of VGP in Cambro-Ordovician Age for Antarctica and the declinations of NRM. 1: Ongul Island (NAGATA and SHIMIZU, 1959); 2 and 3: Ongul Island (NAGATA and YAMA-AI, 1961); 4: Lützow-Holm Bay (KANEOKA *et al.*, 1968); 5: Sør Rondane Mts. (ZIJDERVELD, 1968); 6: Mirnyy Station (MCQUEEN *et al.*, 1972); 7: Taylor Valley (MANZONY and NANNI, 1977); 8–12: this study; 13: Wright and Victoria Valleys (BULL *et al.*, 1962); 14: Wright Valley Ferrar dolerite; (FUNAKI, 1982); 15: Wright and Victoria Valleys (Ferrar dolerite; BULL *et al.*, 1962). Equal-area projection.

basement complex of the Wright Valley records the variations of the geomagnetic field during 20–30 m.y. in Cambro-Ordovician. The obtianed VGP positions from granitic rock, Vanda lamprophyre and porphyry, after thermal demagnetization at 500°C, are located near the equator of Africa on the present globe. However, these positions cannot be distinguished definitely from each other, at the 95% level of confidence (see Table 3). The deviation angles (Θ) among these mean NRM's directions are within 8.8° which is of the same order of magnitude as the α_{95} values of the various rock formations. Because the Vanda lamprophyre and porphyry are included in the same age of Victoria Intrusives, (MCKELVEY and WEBB, 1967), those two data are combined in Table 4. This experimental result suggests that the geomagnetic field in Cambro-Ordovician period (500 m.y. ago), or that the whole area was reheated during the last phase of the Victoria

Intrusives, in the ordovician (470–480 m.y. ago).

There are only a limited number of paleomagnetic results for Antarctica and these have been subdivided into East and West Antarctica by geological, geomorphological and geophysical evidences. Whereas East Antarctica is clearly a large continental block and was a part of Gondwanaland, it appears that West Antarctica could be a number of isolated regions (e.g., HAMILTON, 1967). Since West Antarctica and the Antarctic peninsula were subject to a different geological history and drift (e.g., SCHARNBERGER and SCHARON, 1982), we will compare the present data only with other paleozoic paleomagnetic data from East Antarctica.

MANZONI and NANNI (1977) obtained a VGP position, for 470 m.y. ago, from lamprophyre dykes (which may be the same sequence as the Vanda lamprophyre) in Taylor Valley (77.64°S, 163.35°E), 20 km south of Wright Valley. Their VGP position is located in the same area as our VGP position. The value Θ of VGP's for the α_{95} value of Taylor Valley lamprophyre ($\alpha_{95}=10.9$), the separate VGP positions are indistinguishable. MCQUEEN *et al.* (1972) reported a VGP position from charnockitic rocks of Cambro-ordovician age (502 ± 24 m.y.) from Mirnyy Station (66°33′S, 93°01′E). The value, shown in Table 4, is located in almost the same position as the VGP's from Wight and Taylor Valleys.

Along most of the coast line of Lützow-Holm Bay, including Syowa Station (69°01′S, 35°35′E), high-grade metamorphic rocks having pegmatite and metabasite dykes are distributed. (e.g., TATSUMI and KIKUCHI, 1959). Geochronological investigations by Pb/U and Rb/Sr radiometric methods show that most of rock formations have a metamorphic age around 500 m.y., namely are of Cambro-Ordovician Age. Paleomagnetic investigations of these formations have been carried out by NAGATA and SHIMIZU (1959, 1960), NAGATA and YAMA-AI (1961) and KANEOKA *et al.* (1968). They concluded that the Pole of the Earth's magnetic dipole was situated near the equator in Cambro-Ordovician Age, as shown in Table 4 and Fig. 6; VGPs from Wright and Taylor Valleys, from Mirnyy Station and Lützow-Holm Bay are all located between latitudes 1.5°S–21°S and longitudes 17° and 33°E. The mean VGP position of these 9 data yields 9.5°S in latitude and 24.5°E in longitude.

However a VGP position of the same age from Sør Rondane Mountains (72°S, 24°E; ZIJDERVELD, 1968) was separated from the other VGP's from East Antarctica, as shown in Fig. 6. It cannot be concluded that this deviation of the VGP for Sør Rondane Mountains is due to significant geomagnetic secular variation or the influence of non-dipole components of geomagnetic field, the angular deviation Θ is 11.7 between this location and the closest neighboring data (Ongul Island (1) in Table 4) and the values of α_{95} are 7° and 4.5°, respectively.

From these paleomagnetic results we conclude that the VGP relative to East Antarctica, was located south of the equator, in Africa, during the Cambro-Ordovician Age; that Antarctica showed little tectonic motions during 20–30 m.y. in that Age and that the Transantarctic Mountains were included in East

Antarctica.

All of the directions of declination of NRM for Cambro-Ordovician rocks from East Antarctica are shown in Fig. 6. The declination for Ongul Island and Lützow-Holm Bay is adopted from the data by NAGATA and SHIMIZU (1959), because the other studies involved fewer samples, of lesser statistical value. The values of 95% confidence of declination obtained in this study, for granitic rock (α_{95}(dec) = 8.2°) and Vanda lamprophyre and porphyry (α_{95}(dec) = 5.3°), are also illustrated in Fig. 6. The declination directions of the samples of Ongul Island and Sør Rondane Mountains are approximately parallel, as they are also for the samples from Mirnyy Station, Taylor and Wright Valleys. The angular deviation between the two groups is 15°–20° for those two groups. If the Earth's geomagnetic field in the Cambro-Ordovician was similar to that at present, the standard precision of nondipole geomagnetic field at low latitudes in the southern hemisphere should be 8°–10° (COX, 1962). The angular deviation of the two groups of declinations is twice that due to the likely nondipole. If the magnetization age of the two groups of rocks is assumed to be the same, a simple interpretation is as follows: In the Cambro-ordovician Age rock formations in all localities were magnetized in the same direction; afterward East Antarctica was rifted between Queen Maud Land and Wilkes Land. A possibility is that the rifting zone may be along the present Amery Ice Shelf and Lambert Glacier. According to deep seismic soundings and aeromagnetic data by the Soviet Antarctic Expedition in 1973, a crustal feature of the MacRobertson—Princess Elizabeth Land through Lambert Glacier, is characterized by large-scale normal faulting and the resulting development of a deep graben filled with low-density rocks; their density is 2.3 and the country density is 2.8. Beneath the graben, the crustal thickness is as low as 22–24 km, while on both sides of the graben, the crust is 30–34 km thick (KURININ and GRIKUROV, 1982; FEDOROV *et al.*, 1982). From geological evidence those authors inferred that the beginning of rifting was in the late Mesozoic. The topographic features under the ice sheet in those area include a large valley which extends southwards from the Amery Ice Shelf (i.e., FEDOROV *et al.*, 1982). The directions of magnetization of Cambro-Ordovician rocks in East Antarctica support their results completely and suggest a rifting angle of 15°–20°.

The VGP position of granite collected 20 m below the lower boundary of a Ferrar dolerite sill is fairly consistent with that of the Ferrar dolerite obtained by BULL and IRVING (1960), BULL *et al.* (1962) and FUNAKI (1982); their ellipses of 95% confidence overlap each other, as shown in Fig. 6. It seems therefore that the granitic rocks are baked above all Curie points by the intrusion of the dolerite sills. The location of the VGP of Theseus granodiorite collected from bottom of valley is close to that of Ferrar dolerite, but is not consistent completely taking into account α_{95} value. The geological evidence show that the three dolerite sills intruded in Wright Valley, named "a", "b" and "c" from lowest to highest altitude, were not intruded simultaneously, (MCKELVEY and WEBB, 1961). On the other hand there is a possibility that a fourth sill was also intruded

into the basement complex, under the surface of the bottom of the Wright Valley. Although no dolerite sill was observed in a 85.7 m core obtained in Dry Valley Drilling Project (DVDP; 83.6 m in altitude) core No.4 (CARTWRIGHT *et al.*, 1974) at Lake Vanda in the middle of the Wright Valley, a dolerite sill of 39.53 m thickness was observed under 12.6 m of sediments in DVDP core 13 at Don Juan Pond in the upper Wright Valley (MUDREY *et al.*, 1975). Without further information it is different to determine which sills affected the magnetism of the rock formations in the bottom of Wright Valley. Since the thickness of sill "a", "b" and "c" are 250, 180 and 120 m, respectively (MCKELVEY and WEBB, 1961), and the vertical distance from the lower boundary of sill "a" to the bottom of the valley is about 600 m, the thermal influence by sill "a" was the most important of these three for the magnetization of these formations. The mathematical estimation of the temperature in the neighborhood of a cooling intrusive sheet was given by CARSLAW and JAEGER (1959). The assumptions on which the calculation are based, and the notation used, are as follows. The dolerite sill of thickness $2a$ and initial temperature V_1, intrudes into infinite country rock mass of constant temperature V_0. The physical properties of the sill and the country rock are assumed to be the same, thermal conductivity K, density ϱ, specific heat c and thermal diffusivity κ, where $\kappa = K/\varrho c$.
The temperature V, after t time, is given by

$$V = \frac{1}{2}(V_1 - V_0)\left\{\mathrm{erf}\frac{a-x}{2\sqrt{\kappa t}} + \mathrm{erf}\frac{a+x}{2\sqrt{\kappa t}}\right\} \tag{1}$$

where x is the distance from center of sill, and

$$\mathrm{erf}\,x = \frac{2}{\sqrt{\pi}}\int_0^x \mathrm{e}^{-\xi^2} d\xi \tag{2}$$

is the tabulated error function. When the values K and c are assumed to be 0.006, 2.6 and 0.25 respectively, $\kappa \approx 0.01$ in Eq.(1). The relations among V, x and t are shown in Fig. 7, where $V_0 = 0°C$, $V_1 = 1{,}100°C$ and $a = 120$ m (thickness of the sill "a"). The maximum temperature attained in the granite at a distance of 120 m from the contact with a dolerite sheet 240 m thick, would be nearly 300°C after 625 years, and at 600 m from the contact would be 100°C after approximately 1,500 years. From this analysis, there is no possibility of heating the bottom of the valley up to 500°C by the heating from sill "a". Similar results were obtained even with reasonable changes of the physical properties (JAEGER, 1957, 1959). Since the vertical distances between centers of the sills are estimated as 490 m for "a" to "b" and 650 m for "b" to "c", sills "b" and "c" would have almost no thermal effect at the bottom of the Wright Valley. Therefore the most likely thermal source for heating rocks in that location may be a huge buried dolerite body. According to BULL *et al.* (1962), the directions of magnetization of sills "a", "b" and "c" are consistent with each other, at

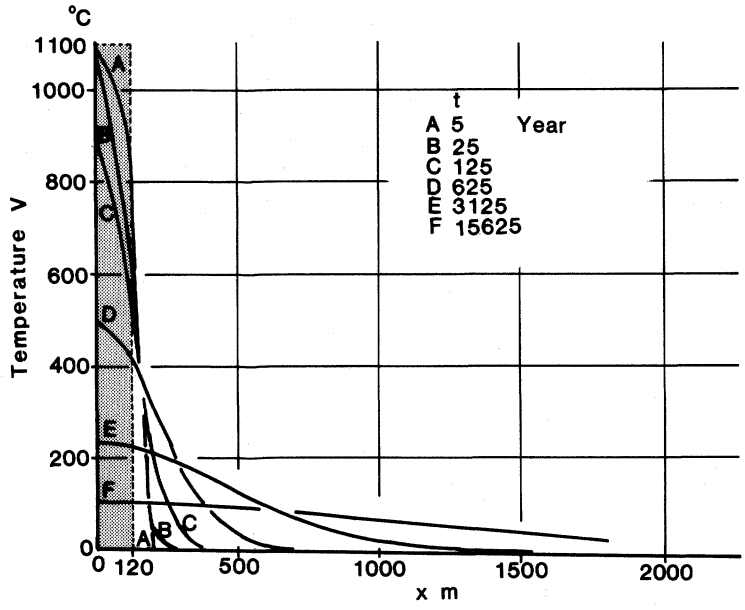


Fig. 7. Temperature in the neighborhood of a Ferrar dolerite sill. Thickness of the sill, $2a=240$ m, temperature of the sill $v_1=1{,}100°C$, thermal conductivity $K=0.01$; distance x m from the center of the sill.

the 95% confidence level. Since the ellipse of α_{95} for Theseus granodiorite does not overlap with them, the rock formations of the valley bottom have a possibility to be heated by hidden sills, rather than by sills "a", "b" and "c". The maximum increased temperature by intrusion for granitic rocks of 20 m below from lower boundary of sill "a" is estimated more than 600°C from Fig. 7. Their Curie points are estimated to be 300 to 400°C, from the thermal demagnetization curves; NRM directions disperse at a temperature of 600°C (see Fig. 5(a)). Therefore we conclude that these granitic samples were heated by sill "a" above the Curie point and were remagnetized on subsequent cooling so that the directions of magnetization of these samples are the same as those of the Ferrar dolerites.

7. Concluding Remarks

The specimens collected from granitic rocks, Theseus granodiorite, Vanda lamprophyre and porphyry in the Wright Valley have a fairly stable magnetization and "red-dyke" rocks have a component of stable magnetization against AF demagnetization. Magnetic grains in these samples apparently have single or pseudosingle domain structure. Almost all specimens are magnetized with normal polarity and their NRM declinations are whithin the 220° and 250° meridians.

Formations at the bottom of the Wright Valley were heated up to 500°C,

probably by the intrusion of a large hidden dolerite body of Jurassic Age. Consequently, the primary magnetizations acquired in Cambro-Ordovician times survived only in those samples which have a Curie point higher than 500°C; the samples which have lower Curie point than that temperature were completely remagnetized in the magnetic field direction of the Jurassic. In the case of granitic rocks, Vanda lamprophyre and porphyry, superposition of the primary and the secondary magnetization is observed at room temperature, but these two kinds of magnetization can be separated by thermal demagnetization at 500°C. On the other hand, samples from Theseus granodiorite and the "red dykes" were remagnetized completely and show only the secondary magnetization direction.

VGP positions for the Cambro-Ordovician, obtained from the primary magnetization of Wright Valley rocks, are located south of the equator, in Africa, on the present globe. These positions are consistent with 7 other VGPs obtained from rocks of the same age elsewhere in East Antarctica. However the directions of the mean declinations from those sites fall into two groups; they are approximately parallel for the samples from Ongul Island, and Sør Rondane Mountains, in one group, and from Mirnyy Station, Taylor Valley and Wright Valley, in the other group. The angular deviation in declination between these two groups is 15°–20°. The most simple interpretation of this deviation is that East Antarctica rifted along a zone through the Amery Ice Shelf and Lambert Glacier, the angular opening being 15°–20°. The limited distributions of these VGPs suggests that East Antarctica experienced very little tectonic movement during a 20–30 m.y. period during Cambro-Ordovician Age, and that the Transantarctic Mountains are a part of the East Antarctic Plate.

The author wishes to thank Professor T. Nagata, Director of the National Institute of Polar Research, for his paleomagnetic suggestions and Dr. Y. Kono, University of Kanazawa, for his geophysical suggestions. The sampling for this research was carried out in the 1977–78 austral summer season, with the support of the U.S. National Science Foundation. The manuscript of this paper was carefully reviewed by Professor C. Bull, during his tenure of a Visiting Professorship at the National Institute of Polar Research.

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